

Curriculum Vitae

ABDULLAH SHADEK FAHIM

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RESEARCH INTERESTS

Machine Learning Surrogate, Data-driven Semiconductor Modeling, Ion Implantation, VLSI and Compound Semiconductor Devices (SiC, GaAs), Molecular Dynamics, AI-driven Wearable Biomedical Instrumentation.

EDUCATIONAL QUALIFICATION

Jashore University of Science and Technology (JUST)

January 2020 – May 2026

Bachelor of Science in Electrical and Electronic Engineering

Jashore, Bangladesh

- **CGPA:** 3.80/4.00 (average 3.93 in last 6 semesters) **Rank:** 2nd of 35
- **Relevant Coursework:** Electrical Engineering Materials, VLSI Design and Technology, Digital Electronics and Logic Design, Nuclear Engineering, Plasma Engineering, Signals and Systems, Digital Signal Processing

STANDARDIZED TEST SCORE

IELTS Academic: Overall 7.5 (Listening: 8.0, Reading: 7.5, Writing: 7.0, Speaking: 7.0) - Test Date: 12 September 2026

PROFESSIONAL EXPERIENCES

Research Assistant

February, 2025 – Present

Interdisciplinary Laboratory of Advanced Materials Processing and Simulation

Jashore, Bangladesh

UNDERGRADUATE THESIS

Thesis Title: A Dual-Stage Hybrid Random Forest Framework for Forward and Inverse Modeling of Ion Implantation in Si, SiC, and GaAs Substrates.

Description: In this thesis, a dual-stage gatekeeper-constrained Random Forest surrogate has been developed for the field of semiconductor ion implantation modeling in Si, SiC, and GaAs across five ions (B, Mg, P, Ar, As). This surrogate predicts nine implantation quantities about 715 times faster than a standard SRIM run, with R^2 above 0.94 for projected range and SRIM-level error against published SIMS data. The framework also offers constraint-based inverse recipe generation for target depths. Therefore, this surrogate could be useful for fast and efficient implantation process design.

Live WebApp: <https://ion-implant-surrogate.onrender.com>

Supervisor: Md. Amzad Hossain, Dr. Eng, Associate Professor, Dept. of EEE, JUST

PUBLICATIONS

I. Journal Papers:

1. **A.S.Fahim**, M.Haque, A.Taher, N.M.Sathi, J.Islam, M.M.Hasan, M.A.Hossain, Predicting Ion Implantation Range, Straggle, and Radiation Damage in Si, SiC, and GaAs using a Gatekeeper-Constrained Random Forest Surrogate, *Nuclear Instruments and Methods in Physics Research Section B, Elsevier (Under Review since September 2026)*
Preprint: <https://doi.org/10.2139/ssrn.7580999>
2. M.Haque, **A.S.Fahim**, S.Mustak, N.M.Sathi, J. Islam, M.Tariquzzaman, M.A.Hossain, Hydrogen Implantation and Diffusion in BCC Tungsten: A Systematic Molecular Dynamics Study over 5–300eV and 300-1500K, *Fusion Engineering and Design, Elsevier (Under Review since May 2026)*
3. N.M.Sathi, M.H.Rocky, N.R.Ritu, **A.S.Fahim**, I.A.Mim, M.Haque, J. Islam, M.Tariquzzaman, M.A.Hossain, A Density Functional Theory Study of Penta-Boron Phosphide (P-BP₅) Nanosheet for (NO, SO₂, NO₂, and O₃) Toxic Gas Sensing, *Solid State Communications, Elsevier (Under Review since June 2026)*
Preprint: <https://doi.org/10.2139/ssrn.7030770>

II. Conference Papers:

1. **A.S.Fahim**, J.Islam, I.A.Mim, N.M.Sathi, CardioPredictor: A Real-time Non-Invasive and Cuffless Cardiovascular Disease Prediction System using 10-Fold Cross Validation on the Framingham Dataset, 2025 International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN), IEEEExplore
DOI: <https://doi.org/10.1109/QPAIN66474.2025.11171821>

RELEVANT PROJECTS AND FIELD VISIT EXPERIENCES

- **Varley Loop Test Kit:** Developed a functional test kit for locating underground cable faults for Electronics Lab.
- **Line Fault Detection Circuit:** Designed a line-to-line fault detection and temperature protection system.
- **Self-Balancing Robot:** Developed a two-wheel self-balancing robot using Arduino Nano and MPU6050, implementing a PID control algorithm for stable, real-time balancing.
- **Industrial Visit at Khulna Power Plant:** Participated in an on-site visit to the 330 MW Dual Fuel Combined Cycle Power Plant and observed practical power generation processes as well as industrial energy infrastructure.
- **Energy Audit and Industrial Analysis:** Conducted an on-site evaluation at Bismillah Foundry and Light Casting Industries Ltd., analyzed electrical energy usage, lighting, HVAC and identified power factor inefficiencies.

SOFTWARE AND TECHNICAL KNOWLEDGE

- **Programming Languages:** Python, MATLAB/SIMULINK, C, 8085 and 8086 Assembly Language
- **Engineering Software:** SRIM/TRIM, Materials Studio, Modelsim, Microwind, Multisim, Proteus, PowerFactory, Autocad, EasyEda
- **ML Tools & Frameworks:** Keras, Numpy, Tensorflow, Pandas, Scikit-learn
- **Data Visualization:** Matplotlib, Origin Pro
- **Hardware Knowledge:** Arduino, ESP8266, ESP32, MPU6050, PLC, DCS
- **Other Software:** Microsoft office, Adobe Illustrator, Adobe Photoshop, Adobe Indesign

TRAINING, WORKSHOP AND CERTIFICATIONS

- Training on *Solar System Planning, Design, Installation, and Operation for Engineering Students* by Daffodil International University (Certificate: <https://lnkd.in/g6mbTzWu>)
- Industrial Training on *Analog IC design, custom layout methodology, and place-and-route in an industry* by Ulkasemi Pvt. (Certificate: <https://lnkd.in/gRDDibeZ>)
- 2025 Nuclear Engineering Summer School (NESS) by the Consortium for Monitoring, Technology, and Verification (MTV) (Certificate: <https://lnkd.in/g4nVMutU>)
- Training on *Industrial Technology on Electrical Engineering and Instrumentation* by Training Institute for Chemical Industries (TICI) (Certificate: <https://lnkd.in/gdvD9YNK>)

LEADERSHIP AND VOLUNTEERING

- **Mentor, IEEE JUST Student Branch** **December 2025 – Present**
- **Chairperson, IEEE JUST Student Branch** **September 2024 – December 2025**
 - Led all strategic and technical initiatives, drove membership growth, and initiated the IEEE JUST Signal Processing Society and Women in Engineering (WIE) Affinity Group.
- **Lead Volunteer (Technical), 5th IEEE International Conference on Robotics, Automation, Artificial-intelligence and Internet-of-Things (RAAICON 2026), IEEE Bangladesh Section**
- **Volunteer, 27th IEEE International Conference on Computer and Information Technology (ICCIT 2024), IEEE Bangladesh Section**

HONORS AND ACHIEVEMENTS

- **Outstanding Student Volunteer Award (Honorable Mention) 2025** by IEEE Bangladesh Section.
- **Second Runner-Up** at IEEE Bangladesh Section SYW Idea Pitching Competition
- **Keynote Speaker – From Idea to Publication: Writing a Conference Paper**, IEEE JUST Student Branch
- **Junior School General Scholarship 2014**, Board of Intermediate and Secondary Education, Chittagong, Bangladesh

ADDITIONAL EXPERIENCE

Fiverr International Ltd.
Professional in book formatting, graphic design and website design (350 completed projects)

June 2021 – Present

REFERENCES

Md. Amzad Hossain, Dr. Eng. Associate Professor, Dept. of EEE Jashore University of Science and Technology (JUST) Email: ma.hossain@just.edu.bd Relationship: Academic Supervisor	Dr. Mehedi Hassan (Jewel) Associate Professor, Dept. of EEE Jashore University of Science and Technology (JUST) Email: m.hassan@just.edu.bd Relationship: Academic Advisor	Professor Dr. Engr. Imran Khan Chairman, Dept. of EEE & Dean, Faculty of Engineering and Technology, JUST Email: I.khan@just.edu.bd Relationship: Head of Department
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